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Task Description:

The project is organized as two Specific Aims: (1) Engineer functional human skeletal tissue, and (2) Engineer functional cardiac tissue comprising a thick assembly of viable cells. The global hypothesis is that biophysical stimuli that are physiologic in nature, magnitude and regime of application can direct cells to regenerate functional tissues in vitro. We can test this hypothesis in controlled in vitro studies of cells cultured on 3D biomaterial scaffolds in advanced bioreactor systems. Our studies of engineered skeletal and cardiac tissue constructs involve the cultivation of cells on 3D biomaterial scaffolds in advanced bioreactor systems in simulated microgravity (rotating vessels) and in the presence of biophysical stimuli. Critical research areas include cell sourcing (stem cell differentiation, expansion, co-culture), cell seeding of scaffolds, construct cultivation in vitro for controlled studies of biophysical stimuli (e.g. hydrodynamic shear, cyclic compression or stretch), and in vivo studies (e.g. construct survival, integration, capacity for defect repair). Tissue engineered constructs are assessed at timed intervals with respect to structure, composition, and mechanical function to evaluate individual and interactive effects of experimental variables. Data is analyzed in a stratified manner and utilized to tailor model system parameters such that engineered tissue properties approach those of native tissues. Construct properties are correlated with: (i) in vitro culture conditions, and (ii) the outcome of in vivo studies, in order to provide feedback necessary to tailor model system parameters and obtain the best overall result. The expected outcomes are to help define how biophysical stimuli regulate skeletal and cardiac tissue development and function, and to yield engineered skeletal and cardiac tissues with superior functionality to that previously reported in the literature. Our interdisciplinary NASA research project is devoted to engineering functional skeletal and cardiac tissues, in vitro for studies on earth and in space, and in vivo for tissue repair. We selected to study skeletal and cardiac tissues because: (a) spaceflight adversely affects these tissue types, (b) improved treatments are needed for millions of patients who suffer from progressive cartilage and heart diseases, (c) these two distinctly different tissue types can provide paradigms for studies of individual and interactive effects of biophysical stimuli, (d) we and others have already demonstrated the feasibility of engineering skeletal and cardiac-like tissue, and thus can capitalize on recent advances to promote groundbreaking research and technology transfer. We previously designed an in vitro system based on cells, three-dimensional (3D) scaffolds and advanced bioreactor systems, and demonstrated that tissues engineered in bioreactors were larger and functionally superior to controls generated in petri dishes. We characterized the tissues by quantitative structural, molecular, and functional assays, and showed that tissue properties depended on model system parameters in ways that could be mathematically modeled. The current limitations of our ongoing microgravity tissue engineering project are as follows: (i) functional engineered cartilage based on human cells has not yet been made in vitro, and incomplete integration of engineered cartilage with adjacent host tissues may limit its long-term mechanical properties, (ii) engineered cardiac tissue has only been made in very thin layers and from non-human cells that are present at low number and low viability, and (iii) spaceflight tissue culture studies are limited by the lack of proper controls necessary to derive underlying mechanisms. Objectives of the current project include: (a) integrated studies in cell and molecular biology, mass and momentum transport, bioreactor design and operation, tissue biomechanics, implant biocompatibility and remodeling, and microgravity research, (b) experiments interpreted with the aid of mathematical models, (c) complementary in vitro and in vivo studies, and (d) functional evaluation at various hierarchical scales, i.e. from molecular to cell and tissue levels. The current project is being realized through the joint efforts of researchers located at the Massachusetts Institute of Technology (MIT) (Aims 1, 2, and 3) and Duke University (Aims 1 and 2), with collaborators at Harvard Medical School (Aims 1 and 2). Our research project is relevant to Critical Path Roadmap (CPR) questions http://Criticalpath.jsc.nasa.gov/ #1) bone loss, #2) impaired fracture healing, #3) injury to joints, #6) diminished cardiac function, #13) skeletal muscle atrophy, and #23) return to gravity/rehabilitation. Our combined use of engineered skeletal and cardiac muscle tissue constructs and advanced bioreactors provides a controlled in vitro environment for studies of significance to NASA's mission. Biophysical stimuli play a critical role in growth, maintenance, and remodeling of musculoskeletal and cardiovascular tissues. During human spaceflight, altered biophysical stimuli and the resulting systemic adaptations cause fluid shifts, skeletal and cardiac deconditioning, and loss of bone and muscle mass. Exercise regimens and pharmacologic interventions have been proposed to try to maintain the health of skeletal and cardiac tissues in-flight, and to try to rehabilitate deconditioned skeletal and cardiac tissues post-flight. However, the ability of existing countermeasures can actually maintain astronaut health is largely unknown. Our controlled studies, in high fidelity in vitro models, of how skeletal and cardiac tissues respond to simulated microgravity and/or biophysical stimuli are expected to help identify mechanisms by which the microgravity environment of space may compromise the ability of skeletal and cardiac tissues to respond to and/or withstand biophysical stimuli. Improved understanding is expected to aid in the development of better countermeasures and is thus relevant to the NASA theme of Human Health and Countermeasures (HH&C) and to human exploration of space. Microgravity tissue engineering research has implications not only for astronaut health but also for ground-based medicine, e.g. obesity related degenerative joint diseases and hypertensive cardiomyopathy. Skeletal and cardiac tissues perform structural and mechanical functions vital for health and survival. Incidences of osteoarthritis and heart failure are increasing and, in the absence of curative interventions, have resulted in substantial human suffering and medical expense. Over forty million Americans suffer from arthritis (www.aaos.org), which is characterized by the degeneration of cartilage and results in chronic pain and progressive disability. Once damaged, cartilage has a limited capacity for self-repair, and no currently available therapy can predictably restore a functional articular surface. Over five million Americans suffer from heart failure, which is characterized by the degeneration of cardiac muscle and is a leading cause of morbidity and mortality in developed countries. The only currently available definitive treatment for end stage heart failure is cardiac transplantation, but limited organ availability leads to long waiting periods that are often not survivable. Tissue engineering is emerging as a significant clinical option to address tissue and organ failure by implanting a cell-based tissue substitute that can provide immediate functionality and the capacity to grow and integrate with surrounding host tissues. However, although cell-based procedures for articular cartilage repair procedures are already in widespread clinical use, critics maintain that the field is principally empirical in nature, and the results unsatisfactory with respect to integrative repair and durability. Significant challenges also remain to be solved before cell-based procedures for cardiac repair can be of clinical use, especially in the areas of cell sourcing and construct biomechanical function. Microgravity tissue engineering studies are thus expected to be of critical importance both for the human exploration of space and for ground-based medicine.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	Skeletal and cardiac tissues perform structural and mechanical functions vital for health and survival. Incidences of osteoarthritis and heart failure are increasing and, in the absence of curative interventions, have resulted in substantial human suffering and medical expense. Over forty million Americans suffer from arthritis (www.aaos.org), which is characterized by the degeneration of cartilage and results in chronic pain and progressive disability. Once damaged, cartilage has a limited capacity for self-repair, and no currently available therapy can predictably restore a functional articular surface. Over five million Americans suffer from heart failure, which is characterized by the degeneration of cardiac muscle and is a leading cause of morbidity and mortality in developed countries. The only currently available definitive treatment for end stage heart failure is cardiac transplantation, but limited organ availability leads to long waiting periods that are often not survivable. Tissue engineering is emerging as a significant clinical option to address tissue and organ failure by implanting a cell-based tissue substitute that can provide immediate functionality and the capacity to grow and integrate with surrounding host tissues. However, although cell-based procedures for articular cartilage repair procedures are already in widespread clinical use, critics maintain that the field is principally empirical in nature, and the results unsatisfactory with respect to integrative repair and durability. Significant challenges also remain to be solved before cell-based procedures for cardiac repair can be of clinical use, especially in the areas of cell sourcing and construct biomechanical function. Microgravity Tissue Engineering studies are thus expected to be of critical importance not only for the human exploration of space but also for ground-based medicine.
	Our skeletal tissue engineering studies (Aim 1) involve the cultivation of cells on various 3D scaffolds in bioreactors in the presence of biophysical signals. We used chondrocytes and polyglycolic acid non-woven mesh to generate cartilage constructs with mechanical properties suitable for in vitro loading studies, and then studied the effects of cyclic compressive loading in bioreactors designed to simultaneously provide perfusion of culture media (Seidel, Biorheology, 2004). Each discoid construct was positioned below a mobile piston in a perfused chamber, and various regimens of unconfined compressive loading was applied for up to 10-weeks. Final constructs contained higher amounts of glycosaminoglycans (GAG, a key component of cartilage) at their top surfaces (the area in opposed to the compression plate) and lower amounts at their radial surfaces. These data implied that unconfined cyclic compression induced GAG to leach out of the construct via its radial surfaces. GAG leaching, which was more pronounced at higher stimulation frequencies and with longer periods of mechanical stimulation, was attributed to the structure and composition of the scaffold. We next explored a novel scaffold: a porous sponge made of biodegradable silk modified with the oligopeptide ligand RGD (arginine-glycine-aspartic acid) that binds many integrin receptors (in collaboration with Prof. D. Kaplan of Tufts University). Sponges made of unmodified silk, collagen, and cross-linked collagen served as controls. The scaffolds were seeded with human mesenchymal stem cells (hMSC) harvested from adult human bone marrow and evaluated with respect to their ability to undergo chondrogenesis (Meinel, Biotechnol. Bioeng, 2004) and osteogenesis (Meinel, Ann Biomed Eng, 2004; Meinel, J Biomed Mater Res, 2004) over a period of 4-5 weeks in vitro. At timed intervals, constructs were sampled to evaluate histological structure and biochemical and molecular components; bone tissue development was also monitored non-destructively by microcomputerized tomography. The hMSC formed cartilaginous tissues when cultured in media with chondrogenic additives. The deposition of GAG and the amounts of cartilage specific type II collagen and its mRNA were all higher for hMSC cultured on silk than on collagen scaffolds. The silk sponge was thus an appropriate scaffold for engineering cartilage starting from hMSC, a finding attributable to its structural integrity, high porosity and slow degradation rate. The hMSC formed bone-like tissue when cultured in osteogenic media. Bone-like trabeculae that were over one millimeter long developed on silk-RGD scaffolds and to a lesser degree on unmodified silk, but were not present on collagen scaffolds. The X-ray diffraction pattern of the deposited bone corresponded to that of the hydroxyapatite present in the native bone. Constructs based on silk-RGD scaffolds also contained more mineralized matrix, osteopontin, alkaline phosphatase, and bone sialoprotein than constructs based on unmodified silk or collagen scaffolds. In spinner flask cultures, medium flow at the construct surfaces was associated with the formation of bone rods that were interconnected, located at the construct periphery, and oriented perpendicular to the construct surface. In contrast, in perfused cultures, medium flow through the construct interior resulted in the formation of bone rods that were present throughout the construct and oriented in the direction of fluid flow. Together, these results suggest that engineered cartilage and bone can be generated starting from human stem cells, 3D scaffolds, and bioreactors, and that chondrogenesis and osteogenesis can be modulated by the properties of the 3D scaffold (material, structure, and degradation rate) and the flow environment (mass transport, compression, and hydrodynamic shear). We previously utilized engineered cartilage in conjunction with a subchondral support material to study the integrative repair of large osteochondral defects in adult rabbits over a period of 6 months (Schaefer, Arthritis & Rheumatism, 2002). In a follow-up study (Schaefer, Orthopade, 2004), we evaluated the pattern of new bone formation by assessing the incorporation of three fluorochromes that had been administered sequentially during the post-operative period: (i) xylenol orange at postoperative week 2, (ii) calcein green at postoperative week 5, and (iii) oxytetracycline at postoperative week 13. We found that new bone (osteoid) was deposited in a linear pattern, and that extensive remodeling was evident in the subchondral region. We concluded that composite implants integrated well with the host bone in 90% of the cases, but did not integrate well with the host cartilage, consistent with previous studies showing that the functional integration of engineered cartilage is a problem of major clinical relevance. We next carried out controlled studies of engineered cartilage integration with adjacent tissues, utilizing rotating bioreactors as the in vitro model (Tognana, Osteoarthritis Cartilage, in press). We tested the hypothesis that time and adjacent tissue properties affected integrative repair of engineered cartilage. Engineered constructs made of chondrocytes and hyaluronan benzyl ester mesh were placed in adjacent tissue rings made of articular cartilage, devitalized bone, or vital bone and cultured for up to 8 weeks. Significant independent and interactive effects of time in culture and type of adjacent tissue on construct adhesive strength, compressive modulus (stiffness), GAG and collagen contents, and cellularity were observed. For constructs cultured in rings of cartilage, negative correlations were observed between ring GAG content and construct properties. Integrative repair was better for constructs cultured adjacent to bone than cartilage, in association with its solid architectural structure and high GAG content, and best for constructs cultured adjacent to DB, in association with its high diffusivity. We concluded that integrative repair in engineered cartilage improved with time and depended on adjacent tissue architecture, composition, and transport properties. We subsequently studied the functional integration of engineered cartilage constructs in an ectopic in vivo model (Tognana, Trans ORS, 2005). In particular, we used subcutaneous implants in immunodeficient (nude) mice to study construct integration with native or enzymatically treated cartilage, and with native or devitalized bone. We found significant effects of implant duration and adjacent tissue structure and composition on construct properties that were qualitatively similar to our in vitro study. In a related study (Leddy, J Biomed Mater Res, 2004), we characterized the transport properties in engineered cartilage in vitro. The development of experimental and theoretical methods to assess the diffusion of marker and biologically active molecules in engineered tissue constructs is important because diffusion is likely to be the primary mechanism for macromolecular transport in the constructs, and because cell proliferation and extracellular matrix deposition requires an adequate supply of nutrients and growth factors. We studied stem cells

Task Progress:

derived from human adipose tissue in conjunction with four types of scaffolds (alginate, agarose, gelatin, or fibrin) and two types of culture media (with or without chondrogenic additives). Diffusion coefficients of four different-sized fluorescently labeled dextran markers were assessed by fluorescence recovery after photobleaching. The constructs exhibited much higher diffusivities than native cartilage. Over 1 month of in vitro culture, construct diffusivity decreased markedly and in a manner enhanced by chondrogenic media. The decrease in construct diffusivity with time in culture was attributed to cell-mediated synthesis and contraction of the extracellular matrix. In contrast, the diffusivity of cell-free scaffolds increased significantly with time in culture, due to degradation and swelling. Our cardiac tissue engineering studies (Aim 2) involve the cultivation of heart cells on various 3D scaffolds in bioreactors in the presence of biophysical signals. In one study (Boublik, in review), we generated hybrid cardiac constructs with mechanical properties suitable for in vitro loading studies and in vivo implantation starting from neonatal rat heart cells, fibrin, and a biodegradable knitted fabric made of hyaluronan benzyl ester. Constructs were compared to native heart tissue, studied in vitro with respect to mechanical function (e.g. stiffness, ultimate tensile strength, failure strain) and compositional remodeling (collagen, DNA), and implanted in vivo. Initially, construct stiffness was mainly determined by the fibrin component and was half as high as native heart, whereas ultimate tensile strength and failure strain were determined by the knitted fabric and were significantly higher than native heart. Over 1 week of in vitro culture, cell-mediated, serum-dependent remodeling was demonstrated by an increase in construct collagen content and maintenance of stiffness not observed in cell-free constructs. Cyclic stretch further increased construct collagen in a manner dependent on loading regime. Presence of cardiac cells in cultured constructs was shown. However, in vitro culture reduced the mechanical properties of the kntted fabric and therefore of cultured constructs. We subsequently studied the development and functional integration of freshly seeded cardiac constructs in an ectopic in vivo model: subcutaneous implants in immunodeficient (nude) mice for up to 3 weeks. Immunohistochemical analysis of explants indicated the continued presence of cardiomyocytes and blood vessel in-growth. Together, the data showed that hybrid cardiac constructs initially exhibited appropriate biomechanical properties, and remodeled in response to serum and stretch in vitro and in an ectopic in vivo model. In another study (Radisic, Am J Physiol, 2004), we hypothesized that functional constructs with physiologic cell densities can be engineered in vitro by mimicking convective-diffusive oxygen transport normally present in vivo. To test this hypothesis, we designed an in vitro culture system that maintained efficient oxygen supply to the cells at all times during cell seeding and construct cultivation. Neonatal rat cardiomyocytes were seeded on collagen sponges using Matrigel? as a delivery vehicle, and cultured with interstitial flow of medium for 1 week. Constructs were assessed at timed intervals with respect to cell number, distribution, viability, metabolic activity and cell cycle, presence of contractile proteins (sarcomeric alpha actin, troponin I and tropomyosin) and contractile function in response to electrical pacing (excitation threshold, maximum capture rate). Interstitial flow of culture medium resulted in physiologic density of viable and differentiated cells, whereas dish culture resulted in constructs with only a thin (0.1-0.2 mm) surface layer containing viable and differentiated cells around an acellular interior. Perfusion resulted in significantly higher numbers of live, metabolically active cells, and significantly more cells in the S-phase as compared to control constructs grown in petri dishes. In response to pacing, perfused constructs contracted synchronously, had superior electrophysiological properties, whereas dish-grown constructs exhibited arrhythmic contractile patterns. In a related study (Radisic, in review), we hypothesized that the functional assembly of engineered cardiac muscle can be further enhanced by oxygen supply provided by mechanisms resembling those in normal vascularized tissues. To mimic the capillary network, cardiomyocytes and fibroblasts isolated from the neonatal rat hearts were cultured on a scaffold made of a highly porous elastomer that had a parallel array of channels and was perfused with culture medium. To mimic oxygen supply by hemoglobin, the culture medium was supplemented with a perfluorocarbon (PFC) emulsion; constructs perfused with unsupplemented culture medium served as controls. In PFC-supplemented media, the decrease in the partial pressure of oxygen in the aqueous phase was only half that in unsupplemented media. Consistently, constructs cultivated in PFC-supplemented media contained higher amounts of DNA and cardiac markers, and had better contractile properties as compared to control constructs. Improved properties of cardiac constructs by the combined use of channeled scaffolds and PFC was attributed to enhanced supply of oxygen to the cells. To rationalize the experimental observations for cardiac constructs cultured using channeled scaffolds and oxygen carriers, we developed a steady state mathematical model of oxygen distribution and solved the model using finite element method (Radisic, in review). The construct was divided into an array of cylindrical domains with a channel in the center and tissue space surrounding the channel. In the channel, the main modes of mass transfer were axial convection and radial diffusion. In the tissue region, mass transfer was by axial and radial diffusion, and the consumption of oxygen was by Michaelis-Menten kinetics. Neumann boundary conditions were imposed at the channel center and half distance between the domains. PFC supplementation of culture medium improved mass transport by increasing the convective term and the effective diffusivity of the medium. The model was first implemented for the following set of parameters: construct thickness 2 mm, channel diameter 330 um, channel wall-to-wall spacing 270 um and average linear velocity per channel of 0.049 cm/s, in conjunction with PFC supplemented and non-supplemented (control) culture medium. Subsequently, the model was used to define favorable scaffold geometry and flow conditions necessary to cultivate engineered cardiac constructs of high cell density and clinically relevant thickness. In another study (Radisic, PNAS USA in press), we hypothesized that electrical stimulation that is physiological in nature, amplitude and frequency, is essential for enhancing functional cell assembly in engineered constructs. Neonatal rat heart cells were seeded at densities corresponding to those normally present in the adult rat heart on collagen sponges, using Matrigel® as a vehicle for rapid cell delivery. Constructs were pre-cultured for 3 days in orbitally mixed dishes to allow the cells to attach to scaffolds and to recover from isolation, and then subjected for an additional 5 days to electrical field stimulation designed to induce synchronous construct contractions (continuous trains of supra-threshold rectangular pulses, 2 ms in duration, 5 V amplitude, 1 Hz frequency). Over only 8 days in vitro, electrical field stimulation induced cell alignment and coupling, increased the amplitude of synchronous construct contractions by seven-fold and resulted in a remarkable level of ultrastructural organization. Development of conductive and contractile properties of cardiac constructs was concurrent, with strong dependence on the initiation and duration of electrical stimulation. These data strongly suggest that biomimetic electrical signals have major beneficial effects on the functional assembly of engineered cardiac muscle. In a related study (Pedrotty, in review), we investigated the effects of electrical stimulation on skeletal myoblasts, i.e. immature muscle cells currently being injected into damaged myocardium in attempts to regenerate heart tissue (cardiomyoplasty). We hypothesized that electrical and biochemical aspects of the cardiac milieu in vivo contribute to myoblast proliferation and plasticity. We tested this hypothesis in a novel in vitro system that enabled separation of the electrical and biochemical signals. Myoblasts were grown on polyglycolic acid mesh under control conditions, in the presence of cardiac-like electrical current fluxes, or in the presence of culture medium that had been conditioned by mature cardiomyocytes. Cardiac-like electrical current fluxes and cardiomyocyte-conditioned culture medium caused myoblast proliferation, but did not affect differentiation of immature skeletal myoblasts along skeletal or cardiac lineages. We next carried out pharmacological studies to try to explain the proliferative effect of electrical stimulation. We used verapimil, a L-type calcium channel blocker that prevents calcium fluxes that may be induced by cardiac-like electrical current fluxes. We showed that the proliferative effects of electrical stimulation on

	myoblasts were ablated by verapamil, and hence acted via an L-type calcium channel-dependent pathway. This study was one of the first determine an underlying mechanism that could potentially be used to improve the properties of engineered cardiac tissue, and also provided a platform for dissecting in vitro the effects of multiple complex stimuli that occur in vivo. Exploiting this or a similar model system for the growth of stem cells could be an excellent means to study cell survival and differentiation in the heart.
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